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Department of Health Research
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INDIAN COUNCIL OF MEDICAL RESEARCH
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Standard Treatment Workflow (STW)

POST-ASPHYXIAL MANAGEMENT OF NEONATES

ICD-11: KB21

1. IDENTIFY RISK
Resuscitation · Delayed cry
Acute perinatal event
Abnormal neurological signs

→

2. STABILIZE
Breathing · Perfusion
Temperature · Glucose

→

3. ASSESS
Encephalopathy
(Modified Sarnat staging)

HIGH-RISK FEATURES: Apgar ≤5 at 10 min · Still requiring PPV at 10 min · Cord or within 1 h: pH ≤7.1 or base deficit ≥12 mmol/L

ASSESS FOR HYPOXIC ISCHEMIC ENCEPHALOPATHY (HIE): MODIFIED SARNAT STAGING

DOMAIN	MILD	MODERATE	SEVERE
Consciousness	Hyperalert	Lethargic	Stupor / coma
Spontaneous activity	Normal / increased	Decreased	Absent
Posture	Normal	Distal flexion	Decerebrate
Tone	Normal / increased	Hypotonia	Flaccid
Primitive reflexes	Suck: normal / incomplete Moro: exaggerated	Suck: weak Moro: incomplete	Suck / Moro: absent
Autonomic signs	Pupils: normal; HR: normal / fast Breathing: regular	Pupils: constricted; HR: slow Breathing: periodic / irregular	Pupils: dilated / deviated HR: variable; breathing: apnoea

Record seizures separately; Repeat serial assessments; Initial normal examination does not exclude evolving HIE

NORMAL EXAMINATION

- Breastfeed; Skin-to-skin if stable
- Observe ≥24 h; longer if concerns

MILD HIE

- Supportive care in SNCU/NICU
- Serial neurological assessments
- Seizures/worsening: Urgent review/referral

MODERATE / SEVERE HIE OR UNSTABLE

- Stabilize: Contact referral NICU
- Arrange urgent transfer for advanced care

MONITORING AND ACTION

PARAMETER	FREQUENCY	ACTION
HR / breathing / SpO_2	Continuous; record hourly	SpO_2 target 91–95% on supplemental oxygen Ventilation if apnoea / inadequate breathing
Perfusion / BP	At least hourly initially; more if unstable	Treat shock; avoid indiscriminate fluid boluses
Neurological status	Hourly till 6 h; then 4-hourly for 24–48 h; immediately if worsening	Detect and treat seizures EEG/aEEG where available
Temperature	Hourly until 6 h; 2-hourly once stable	Aim 36.5–37.0°C (unless under therapeutic hypothermia [TH]) Avoid hyperthermia (>37.5°C)
Blood glucose	At 1, 2 and 6 h; then 6-hourly once stable	Aim 50–125 mg/dL; correct abnormalities IV glucose if feeding unsafe
Fluids / urine output	Record intake/output; review balance	Adjust fluids; baseline renal function / electrolytes Repeat tests; blood gas / calcium as indicated
Feeding	Assess alertness, suck–swallow and tolerance	Breast milk when appropriate Oral feeds when alert and safe

SEIZURES: Act immediately
Stabilize airway and breathing → **Phenobarbitone 20 mg/kg IV over 15 min** (**Refer to Seizures STW**)
Check and correct blood glucose and calcium concurrently; urgent NICU consultation / referral

THERAPEUTIC HYPOTHERMIA: TIME-CRITICAL REFERRAL

Potential candidates: GA ≥36 weeks, moderate/severe HIE + evidence of acute perinatal hypoxia → discuss immediately with referral NICU
Confirm strict NNF criteria (scan NNF QR code below); **initiate TH ASAP within 6 h of birth**
TH only in Level III/IV NICUs with adequate intensive care, continuous monitoring and experienced, trained personnel

DISCHARGE : Stable · Feeding safely · Family counselled · High-risk Neurodevelopmental follow-up including hearing assessment arranged

DOs

- Repeat and document neurological assessments
- Maintain oxygenation, perfusion, temperature and glucose within target ranges
- Counsel the family; ensure safe transfer and documented neurodevelopmental follow-up

DON'Ts

- Do not assume an initially normal examination excludes evolving HIE
- Do not give prophylactic anticonvulsants or unnecessary oxygen/fluid boluses
- Do not delay TH referral in potentially eligible babies or start unmonitored cooling

KEY PERFORMANCE INDICATORS (KPIs)

Timely neurological assessment (%) — Target >95%

$$\frac{\text{At-risk neonates with a documented structured neurological assessment within 1 h*}}{\text{All at-risk neonates}} \times 100$$

*Within 1 h of birth (inborn) / admission (outborn)

Timely physiological assessment (%) — Target >95%

$$\frac{\text{Neonates with suspected HIE with complete physiological assessment** within 1 h of admission}}{\text{All admitted neonates with suspected HIE}} \times 100$$

**Temperature, glucose, oxygenation and perfusion documented

RECOGNIZE EVOLVING HIE EARLY • REASSESS SERIALLY • PROVIDE TIMELY SUPPORTIVE CARE

RELATED INFORMATION

Modified Sarnat assessment
Examination demonstration video

Therapeutic hypothermia
NNF India position statement (2021)

ABBREVIATIONS

aEEG: Amplitude-integrated electroencephalography
BP: Blood pressure
EEG: Electroencephalography
HIE: Hypoxic-ischemic encephalopathy
HR: Heart rate
SpO_2: Peripheral oxygen saturation
NICU: Neonatal intensive care unit
PPV: Positive pressure ventilation
SNCU: Special newborn care unit
TH: Therapeutic hypothermia

REFERENCES

1. NNF Working Group. Position statement and guidelines for use of therapeutic hypothermia to treat neonatal hypoxic ischemic encephalopathy in India. October 2021.
2. NSW Agency for Clinical Innovation. Hypoxic ischaemic encephalopathy in newborns: recognition, monitoring and early management. Clinical practice guide. August 2026.

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